THE LOCUS WHERE A MORPHISM BECOMES A LOCAL COMPLETE INTERSECTION IS OPEN

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11856-11875, immediately following the statement at 11835-11854.

Editorial note. Two cross-chapter pointers (Properties of Spaces) plus one same-chapter lemma are recorded in assembly.external_references.

Editorial note. The displayed description of the point set |W| is retained verbatim.

Allowed standard prerequisites (editorial).

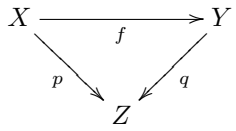
- Base change behaviour of the Koszul locus. chapter lemma lemma-base-change-lci-fibres, cited at proof line 11857. Produces the open subspace U(f) of X where f is Koszul and shows its formation commutes with arbitrary base change.
- Open subspaces of an algebraic space from point sets. Properties of Spaces, lemma spaces-properties-lemma-open-subspaces, cited at proof lines 11862-11863. Realises the constructed set of points W as an open subspace of Z.
- Factoring a morphism through an open subspace. Properties of Spaces, lemma spaces-properties-lemma-factor-through-open-subspace, cited at proof lines 11872-11873. Turns the pointwise description of W into the universal property required by the statement.

Source reference. Original lemma 06CA: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11383–11383. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03IE: focused-seeds/geometry/S9-spaces-where-lci-spaces/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03BZ: focused-seeds/geometry/S9-spaces-where-lci-spaces/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Consider a commutative diagram*



Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

of algebraic spaces. Assume that

(1) *p is flat and locally of finite presentation,*
(2) *p is closed, and*
(3) *q is flat and locally of finite presentation,*

Then there exists an open subspace $W \subset Z$ such that a morphism $Z' \to Z$ factors through W if and only if the base change $f_{Z'} : X_{Z'} \to Y_{Z'}$ is a local complete intersection morphism.

Proof. By Lemma 06CA there exists an open subspace $U(f) \subset X$ which is the set of points where f is Koszul. Moreover, formation of $U(f)$ commutes with arbitrary base change. Let $W \subset Z$ be the open subspace (see Properties of Spaces, Lemma 03BZ) with underlying set of points

$$|W| = |Z| \setminus |p| (|X| \setminus |U(f)|)$$

i.e., $z \in |Z|$ is a point of W if and only if f is Koszul at every point of X above z. Note that this is open because we assumed that p is closed. Since the formation of $U(f)$ commutes with arbitrary base change we immediately see (using Properties of Spaces, Lemma 03IE) that W has the desired universal property. □